EXPERIMENT 16

AIM

Compare a trusted SHA-256 baseline with current files and classify files as unchanged, modified, newly added or missing.

ALGORITHM

1. Define the trusted SHA-256 baseline.

2. Calculate SHA-256 for current files.

3. Compare baseline and current filenames.

4. Report missing files.

5. Report newly added files.

6. Compare hashes for common files.

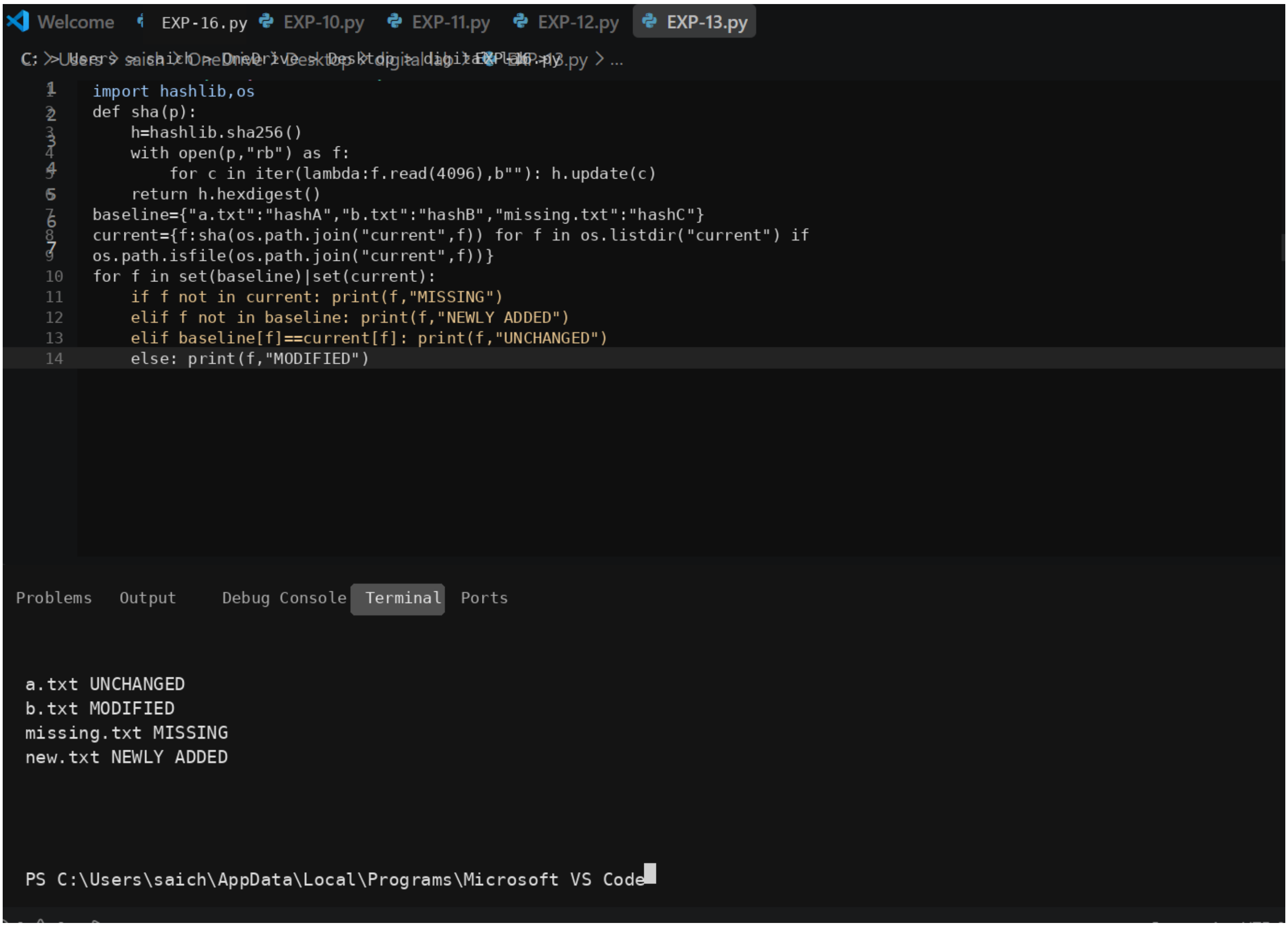
7. Classify them as unchanged or modified.

8. Stop.

CODE

```
import hashlib
import os

def sha256_file(path):
    digest = hashlib.sha256()
    with open(path, "rb") as file:
        for chunk in iter(lambda: file.read(4096), b""):
            digest.update(chunk)
    return digest.hexdigest()

baseline = {
    "a.txt": "hashA",
    "b.txt": "hashB",
    "missing.txt": "hashC"
}

current = {}
for name in os.listdir("current"):
    path = os.path.join("current", name)
    if os.path.isfile(path):
        current[name] = sha256_file(path)

for name in sorted(set(baseline) | set(current)):
    if name not in current:
        print(name, "MISSING")
    elif name not in baseline:
        print(name, "NEWLY ADDED")
    elif baseline[name] == current[name]:
        print(name, "UNCHANGED")
    else:
        print(name, "MODIFIED",
              "Baseline:", baseline[name],
              "Current:", current[name])
```

OUTPUT



RESULT

The baseline comparison classifies files using presence and SHA-256 equality and identifies files requiring investigation.